📌 Concept 1: Men – Days – Work

Formula

Work ∝ Men × Days

For the **same work**

M1 × D1 / W1 = M2 × D2 / W2

Where

- **M** = Number of men
- **D** = Number of days
- **W** = Work done

📚 Questions

Question 1

If **100 cats kill 100 rats in 100 days**, in how many days will **10 cats kill 10 rats**?

Solution

100 cats → 100 rats → 100 days

So

1 cat kills 1 rat in 100 days

Therefore

10 cats kill 10 rats in 100 days

✅ **Answer: 100 days**

Question 2

14 workers make 17 boxes in 6 days.
How many workers are required to make 289 boxes in 42 days?

Solution

14 workers → 17 boxes → 6 days

Using

Workers × Days ∝ Boxes
(14 × 6) / 17 = (W × 42) / 289
84 / 17 = 42W / 289

Cross multiply

84 × 289 = 42W × 17
W = 34

✅ **Answer: 34 workers**

Question 3

How many men are required to plough 100 acres in 10 days if 10 men plough 20 acres in 8 days?

Solution

10 men × 8 days = 80 man-days

For **20 acres**

80 / 20 = 4 man-days per acre

For **100 acres**

100 × 4 = 400 man-days

Let required men = **M**

M × 10 = 400
M = 40

✅ **Answer: 40 men**

📌 **Concept 2: Efficiency**

Efficiency and time are inversely proportional

Efficiency ∝ 1 / Time

If

A is twice efficient as B

Then

A : B = 2 : 1

Time ratio

1 : 2

Question 4

Ajit works three times as fast as Kapil.
If together they finish the work in 12 days, in how many days can Ajit alone finish it?

Solution

Efficiency ratio

Ajit : Kapil = 3 : 1

Total efficiency

4 units

Together work per day

1/12

1 unit efficiency work

(1/12) / 4 = 1/48

Ajit efficiency

3 units

Ajit work per day

3/48 = 1/16

Time taken by Ajit

16 days

✅ **Answer: 16 days**

Question 5

A is twice as efficient as B.
If together they complete the work in 6 days, how long will each take individually?

Solution

Efficiency ratio

A : B = 2 : 1

Total efficiency

3

Together work rate

1/6

1 unit efficiency

(1/6) / 3 = 1/18

A rate

2/18 = 1/9

B rate

1/18

Time taken

A = 9 days
B = 18 days

✅ **Answer: A = 9 days, B = 18 days**

Question 6

A can complete a job in 12 days.
B is 60% more efficient than A.
How many days will B take?

Solution

Efficiency of A

1

Efficiency of B

1.6

Time is inverse of efficiency

B time = 12 / 1.6
= 7.5 days

✅ **Answer: 7.5 days**

Question 7

Efficiency of A is twice B and efficiency of B is thrice C.
If together they complete the work in 10 days, how many days will C alone take?

Solution

Efficiency ratio

A : B : C = 6 : 3 : 1

Total efficiency

10

Together rate

1/10

1 unit efficiency

(1/10) / 10 = 1/100

C rate

1/100

Time taken by C

100 days

✅ **Answer: 100 days**

📌 **Concept 3: Hour Based Work**

Sometimes work depends on **hours per day**.

Total work = Days × Hours per day

Question 8

Radika can finish a work in 6 days working 5 hours a day.
Geeta can finish the same work in 15 days working 3 hours a day.
In how many days will they finish the work if working together for 3 hours per day?

Solution

Radika total hours

6 × 5 = 30 hours

Radika rate

1/30

Geeta total hours

15 × 3 = 45 hours

Geeta rate

1/45

Together per hour

1/30 + 1/45

LCM

90
3/90 + 2/90
= 1/18

Work done per day (3 hours)

3 × 1/18 = 1/6

Days required

$1 \div (1/6) = 6$

✅ **Answer: 6 days**

Question 9

To complete a work, P takes 50% more time than Q.
If together they complete the work in 18 days, how many days will Q alone take?

Solution

Let time taken by Q

x

Time taken by P

$1.5x$

Rates

$Q = 1/x$
$P = 1/(1.5x)$

Together

$1/x + 1/(1.5x)$
$= 1/x + 2/(3x)$
$= 5/(3x)$

Total work in 18 days

$18 \times 5/(3x) = 1$
$90/(3x) = 1$
$x = 30$

✅ **Answer: 30 days**

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